

T0 Theory: The Hubble Constant as a Pure ξ -Ratio

$\hbar H_0 / m_e c^2 = (\pi/2) \cdot \xi^{10}$ — Cosmological and Particle Scale from One Parameter

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March 2026

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Abstract

This document derives a dimensionless ratio for the Hubble constant H_0 within the T0 theory. While Doc. 026 computes H_0 via the Hubble energy $E_H = E_0 \cdot \xi^{41/4}$, it is shown here that the same result can be formulated more cleanly as a pure ratio:

$$\boxed{\frac{\hbar H_0}{m_e c^2} = \frac{\pi}{2} \cdot \xi^{10}} \quad (1)$$

This connects the cosmological scale (H_0) directly with the particle scale (m_e) through an integer power of ξ and a single geometric factor. Numerically, this relation yields

$$H_0^{\text{T0}} \approx 66.5 \text{ km/s/Mpc},$$

in agreement with the Planck value (67.4, deviation +1.3%) and with the T0 value from Doc. 026 (66.2, deviation −0.5%). The ratio is independent of unit conventions and requires no fractal correction, since it describes the ratio of two energy scales.

.1 Starting Point and Motivation

What Doc. 026 Achieves and What Remains Open

Doc. 026 (Geometric Cosmology) derives H_0 from the Hubble energy

$$E_H = E_0 \cdot \xi^{41/4} \quad (2)$$

and obtains $H_0 \approx 66.2$ km/s/Mpc. The result agrees well with the Planck value; however, the exponent $41/4$ raises a legitimate question: it is not an integer fraction with an obvious geometric meaning.

Doc. 026 states: *"The exponent $41/4$ requires a justification from the ξ -field theory."*

The present document answers this question not by relabelling, but through an alternative formulation that makes a deeper structure visible.

The Principle of Pure Ratios

As set out in detail in Doc. 101 (Circularity of the Constants), dimensionless ratios are the actual substance of the T0 theory. As soon as all units are set to 1 ($\hbar = c = 1$), fractal corrections and unit conventions drop out, and only pure powers of ξ remain.

Specifically, for the lepton mass *ratios* (Doc. 006):

$$\frac{m_\mu}{m_e} = \frac{16/5}{4/3} \cdot \xi^{-1/2}, \quad \frac{m_\tau}{m_e} = \frac{25/9}{4/3} \cdot \xi^{-5/6}, \quad (3)$$

and the Koide formula $Q = 2/3$ follows exactly, without K_{frak} being needed. This cleanness occurs *only* for ratios, not for absolute quantities.

The same logic is applied here to H_0 .

.2 Derivation of the Dimensionless Ratio

Numerical Determination of the Exponent

The dimensionless ratio $\hbar H_0 / (m_e c^2)$ is computed for the known measured values of H_0 :

Source	H_0 [km/s/Mpc]	$\hbar H_0 / m_e c^2$	Exponent n
Planck 2018	67.4	2.814×10^{-39}	9.948
T0 (Doc. 026)	66.2	2.764×10^{-39}	9.950
Local (SH0ES)	73.0	3.047×10^{-39}	9.939

Table 1: Dimensionless ratio $\hbar H_0 / m_e c^2$ and the corresponding ξ exponent for various H_0 measurements. The exponent is defined by $\hbar H_0 / m_e c^2 = C \cdot \xi^n$, $C \equiv \pi/2$.

For all cosmologically grounded measurements (Planck, T0), the exponent lies within 0.05% of $n = 10$. The local SH0ES value deviates more strongly, which is consistent with the fact that local measurements can contain systematic biases due to model dependencies (cf. Section .6).

The Geometric Prefactor $\pi/2$

Numerically, the remaining prefactor is:

$$C_{\text{Planck}} = \frac{\hbar H_0^{\text{Planck}}}{m_e c^2 \cdot \xi^{10}} = 1.584 \quad (4)$$

$$C_{\text{T0}} = \frac{\hbar H_0^{\text{T0}}}{m_e c^2 \cdot \xi^{10}} = 1.556 \quad (5)$$

$$\frac{\pi}{2} = 1.5708 \quad (6)$$

$\pi/2$ lies right between the two values and deviates from each by less than 1%. The interpretation:

Central Result

The Hubble constant satisfies the dimensionless ratio

$$\frac{\hbar H_0}{m_e c^2} = \frac{\pi}{2} \cdot \xi^{10} \quad (7)$$

with an accuracy of $< 1\%$ for all cosmologically grounded measurements.

.3 Why This Finding Should Not Be Dismissed

The Structural Argument

A random system does not produce approximations – it produces noise. The parameter ξ was originally developed from tetrahedral geometry and used primarily for particle masses (Doc. 009). That the same parameter connects the cosmological length scale c/H_0 with the particle scale $\hbar/(m_e c)$ through a simple power ξ^{10} is not a trivial result.

For comparison, the characteristic T0 length scale is

$$L_\xi = \xi \cdot \ell_P \approx 5.39 \times 10^{-39} \text{ m}, \quad (8)$$

and the Hubble length is

$$L_H = c/H_0 \approx 1.37 \times 10^{26} \text{ m}. \quad (9)$$

The ratio is $L_H/L_\xi \approx 2.54 \times 10^{64}$. That this enormous ratio can be expressed in simple powers of ξ through

$$\frac{L_H}{L_\xi} \sim \frac{1}{\xi^{10}} \cdot \frac{\text{geometric factors}}{1} \quad (10)$$

points to a genuine connection between the scales.

Complexity Is Not a Reason for Rejection

In particle physics (Doc. 006), the lepton masses show sub-percent agreement, because the underlying interactions are simple enough to be treated analytically. In cosmology, additional factors play a role:

- Recursive back-reactions of the ξ field on macroscopic scales
- Cumulative effects of the geometric redshift over cosmological distances (Doc. 026)
- Systematic model dependencies of the H_0 measurements themselves

Under these circumstances, a $< 1\%$ agreement is *stronger* than a $< 1\%$ agreement for a lepton mass. The standard must be appropriate to the domain.

The H_0 Tension as an Internal Argument

The well-known Hubble tension between the local ($H_0 \approx 73$) and the cosmological ($H_0 \approx 67$) value has an internal spread of $\sim 9\%$. Each of the three measurements from Table 1 lies within this uncertainty.

The T0 ratio (7) clearly favours the *low* Planck value. This is consistent with T0 cosmology (static universe, geometric redshift), since the local SHOES value is based on distance-ladder methods that embed expansion assumptions.

.4 Geometric Meaning of the Components

The Exponent 10

In contrast to the fraction $41/4$ from Doc. 026, $n = 10$ is an integer with a possible geometric interpretation:

Interpretation	Value
4 spacetime dimensions $\times$ 2 (time-mass duality) + 2	10
3 spatial directions $\times$ 3 (generations) + 1	10
Number of free parameters in the CKM-PMNS matrix	10

Table 2: Possible interpretations of the exponent $n = 10$. Which of them is fundamental remains open and would have to be investigated in a separate document.

Open Point

The geometric derivation of the exponent $n = 10$ from first principles of the T0 ξ -field theory remains an open task. The present document establishes the ratio numerically and argues for its physical significance, without providing a conclusive justification.

The Factor $\pi/2$

In the T0 theory, the time field $T(x, t)$ describes a wave structure at the Planck scale (Doc. 009). The energy of a harmonic oscillator over half a period gives the factor $\pi/2$. Consistent with this is the torus topology of the T0 time field (Doc. 159), in which the transition from linear to cyclic propagation likewise produces a factor of $\pi/2$.

A complete derivation from the Lagrangian (Doc. 019) would be the next step.

.5 Equivalence with Doc. 026

Relation (7) and the formula from Doc. 026 are numerically equivalent. The connection between the two representations,

$$E_H = E_0 \cdot \xi^{41/4} \quad \longleftrightarrow \quad \frac{\hbar H_0}{m_e c^2} = \frac{\pi}{2} \cdot \xi^{10} \quad (11)$$

follows from $E_0 = 1/\xi$ (in natural units) and $m_e \approx (\pi/2) \cdot \xi^{3/2} \cdot \nu$ (Doc. 006). The relation

$$\xi^{41/4} \cdot E_0 = \xi^{41/4-1} = \xi^{37/4} \quad \text{vs.} \quad \frac{\pi}{2} \cdot m_e \cdot \xi^{10} \sim \frac{\pi}{2} \cdot \xi^{3/2} \cdot \nu \cdot \xi^{10} \quad (12)$$

shows that the equivalence carries within it a relation between E_0 , m_e and ν – the same relation that Doc. 041 expresses as $\nu = (4/3) \cdot \xi_0^{-1/2} \cdot K_{\text{quantum}}$.

Interpretation

The ratio (7) is not a new prediction, but a **cleaner formulation** of the result of Doc. 026. It makes the scale connection between cosmology and particle physics transparent: m_e and H_0 are both manifestations of ξ at different power levels.

.6 Placement within the H0 Tension

The Hubble tension ($H_0 \approx 67$ from the CMB vs. $H_0 \approx 73$ from local methods) is one of the central open questions of modern cosmology. All local methods (Cepheids, type Ia supernovae, strong lenses) presuppose a cosmic distance ladder that is based on the Λ CDM expansion model.

In the T0 theory there is no expansion. The observed redshift is a geometric effect of the static T0 vacuum (Doc. 026). This means:

- The Planck value (67.4) comes from the CMB analysis, which also uses Λ CDM assumptions, but is based primarily on geometric angular scales, which are less dependent on expansion.
- The SH0ES value (73.0) is embedded more deeply in expansion assumptions and should, from the T0 perspective, be *systematically* too high.

The ratio (7) favours $H_0 \approx 66.5$ and thus lies 1.3% below the Planck value – within the cosmological measurement accuracy and far from the local measurement. This is a *qualitative* test of T0 cosmology: if the theory is correct, the *true* H_0 value should lie closer to the Planck value than to the SH0ES value.

Qualitative Prediction

The T0 ratio formula predicts that the fundamental H_0 value lies in the range 65–67 km/s/Mpc. Local measurements that systematically yield higher values are to be interpreted as artefacts of the expansion assumption – not as a contradiction of the T0 theory.

.7 Summary

Results of This Document

1. The dimensionless ratio $\hbar H_0 / m_e c^2 = (\pi/2) \cdot \xi^{10}$ connects cosmological and particle scales through an integer power of ξ .

2. The exponent 10 replaces the fraction $41/4$ from Doc. 026 and points to a more fundamental description.
3. No fractal correction factor is needed, since this is a pure energy ratio.
4. The relation favours the low CMB-based H_0 value and is consistent with the static T0 universe.
5. The geometric justification of $n = 10$ and $\pi/2$ from first principles remains an open but clearly defined research task.

Related documents: Doc. 009 (origin of ξ), Doc. 025/026 (T0 cosmology), Doc. 027 (MNRAS analysis), Doc. 006 (particle masses), Doc. 041 (parameter derivation), Doc. 101 (circularity of the constants), Doc. 159 (torus topology).